

Lab 1 - PatientPath Descriptive Paper

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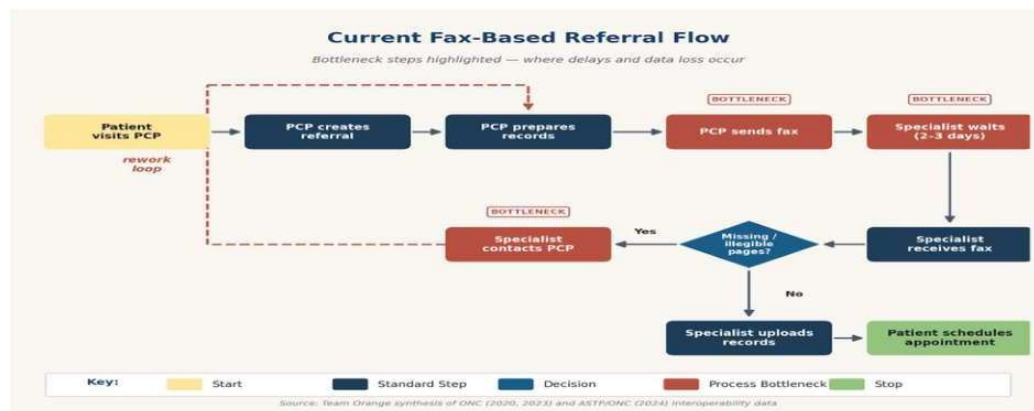
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1 Introduction

Even with the technology available today, healthcare organizations in the United States still have difficulty sharing patient information quickly and consistently. Many providers continue to depend on older methods of communication, which can slow down care, add unnecessary administrative work, and make it harder for clinicians to get the information they need. About 65% of medical record transfers are still completed by fax (ONC, 2023). Electronic exchange is also far from universal. Only 43% of hospitals routinely participate in all four major areas of interoperability: sending, receiving, finding, and integrating electronic health information (ASTP/ONC, 2024). The problem becomes even more noticeable when records have to move outside of a hospital's normal network. Only 16% of hospitals regularly exchange records with outside long-term or post-acute care providers, and only 17% regularly exchange information with behavioral health providers (ASTP/ONC, 2024). Office-based physicians face similar barriers. Eighty-five percent report difficulty exchanging records between different EHR vendors, while 71% work with referral partners that do not have electronic exchange capabilities at all (ONC, 2022). Together, these issues leave patient information spread across systems that often do not communicate well with one another.

Figure 1*Fax Flow*

These gaps translate into real harm for healthcare providers, their support staff, and their patients. Time sensitive referrals stall while staff resend, have to send multiple requests, and manually track status of paperwork (ONC, 2020). Handoffs between the primary care and specialists are incomplete, and referral loops frequently fail to close (Patel et al., 2018). Missing

These gaps have real effects on providers, office staff, and patients. Time-sensitive referrals can be delayed when staff members have to resend documents, submit repeated requests, or manually check whether paperwork was received (ONC, 2020). Communication between primary care providers and specialists can also break down, causing referral loops to remain incomplete (Patel et al., 2018). When important information is missing, patients may undergo repeat diagnostic tests, duplicate imaging, or additional laboratory work that could have been avoided (De Swart et al., 2023). This increases costs while also delaying treatment. Rural and lower-resourced organizations are especially affected because they are less likely to participate in routine interoperability and may also have limited IT infrastructure or unreliable internet access (ASTP/ONC, 2024). Because of this, simply converting a paper process into a digital one is not enough. A useful solution also needs to work with existing EHR systems, protect patient information, remain usable in real-world network conditions, and support HIPAA

requirements.

Figure 2

Impact



For a record-sharing platform to address the problem as a whole, it must account for several connected challenges rather than solving only one part of the process. The major concerns include inefficient record transfers, poor communication between separate systems, privacy and security risks, difficult user experiences, limited scalability, and unreliable connectivity in some healthcare settings.

A system that addresses only some of these problems can still leave important groups behind. For example, a platform may support interoperability but provide little benefit to a rural clinic if it becomes unusable whenever the internet connection is unstable.

PatientPath is the proposed solution to these problems. It is a secure, centralized, web-based platform designed to improve medical record sharing and referral coordination without requiring healthcare organizations to replace the EHR systems they already use. PatientPath uses an API-first design with FHIR R4 and HL7 compatibility to support digital record exchange between different systems. It also combines referral tracking, role-based access control, encryption, patient consent controls, and offline synchronization for locations with unreliable connectivity. The platform is intended for hospitals, private practices, specialists, rural providers, administrative staff, and patients. PatientPath is designed with HIPAA-aligned security controls from the beginning. The following sections describe the product's main features, technical architecture, and case study.

2 Product Description

PatientPath is a secure web platform created to replace slow, fax-based medical record transfers with digital record sharing and coordinated referral management. Instead of asking healthcare organizations to replace their current EHR software, PatientPath works as a connection between existing systems. Records can be received from an organization's EHR, converted into a consistent FHIR R4 format, and sent to another organization through a shared API gateway. Providers, administrative employees, and patients use interfaces designed for their individual roles, while record activity is encrypted, logged, and connected to patient consent.

The main purpose of PatientPath is to reduce the delays, lost documents, and repetitive administrative work that often come with fax-based referrals. At the same time, the platform is intended to improve the privacy and security of record exchange. Its major goals are to support real-time record exchange between organizations using FHIR R4 and HL7 v2.x, provide a single referral workflow with automatic status tracking and escalation, protect access through

encryption, role-based permissions, multi-factor authentication, and audit logs, remain functional in low-bandwidth environments through caching and synchronization, and support reliable cloud deployment with a target uptime of 99.9%.

2.1 Key Product Features and Capabilities

PatientPath combines several features that work together to replace the current fax-based process. Some features are directly visible to users, such as record uploads, referral tracking, consent management, transfer status updates, multi-factor authentication, and offline access. Other features operate behind the scenes, including FHIR/HL7 translation, EHR adapters, encryption, role-based access control, audit logging, and cloud infrastructure.

PatientPath provides secure record uploads by checking files for the expected format and required information before storing them. Records are encrypted with AES-256 using a cloud key management service, with a target upload time of five seconds or less. The platform also supports real-time record sharing. When a referral is created, the receiving organization is given a limited access token for the appropriate records, allowing provider and specialist dashboards to update without requiring staff to manually resend the same information.

PatientPath also provides automated referral tracking. Referrals move through the statuses Pending, In Review, Accepted, and Complete. If no activity occurs for 48 hours, the system creates an administrative alert, followed by supervisor escalation at 72 hours. Patients can also manage consent through the patient portal by granting or removing permission to access their records. When consent is revoked, any active access tokens associated with that permission are also revoked.

Security is supported through multi-factor authentication and role-based access control. PatientPath's real-world design requires TOTP-based two-step verification during login and includes an account lockout and administrative alert after repeated failed login attempts. In the

current prototype, Passport.js sessions and TOTP authentication are being implemented for all users, while the automated lockout and alert process is deferred. PatientPath's real-world design also uses AES-256 encryption for stored information and TLS 1.3 for information in transit. API requests are checked against four defined user roles to ensure that users can only access information permitted by their role.

PatientPath's real-world design includes offline synchronization to support organizations with unreliable connectivity. For the current prototype, this capability has been reduced to read-only caching through a Service Worker. Full queued-write synchronization and conflict resolution are deferred beyond the current prototype scope. The real-world design also includes an interoperability layer that converts information from different EHR systems into FHIR R4 and supports HL7 v2.x messaging for older systems.

PatientPath is different from many existing approaches because it combines three important ideas in one platform. First, it is EHR-agnostic, meaning it is designed to connect existing systems instead of forcing an organization to migrate to a completely new EHR. Second, its offline synchronization is intended to keep the platform useful for rural and low-bandwidth clinics. Third, the single-form referral process connects referral creation, patient consent, record access, and status tracking so that staff do not have to manage the same referral through several disconnected steps.

2.2 Major Components (Hardware/Software)

PatientPath's real-world product design does not require customers to purchase dedicated hardware. Full production deployment is intended to use cloud infrastructure and may include load balancing, replicated databases, encrypted document storage, and cloud-based key management. These production-level infrastructure components are not implemented in the current CS 411 prototype. The prototype instead uses a React front end, an Express API, and

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PostgreSQL 17 running locally through Docker Compose. This reduced architecture allows the team to demonstrate PatientPath's primary functionality within the semester while preserving the larger architecture as a future deployment goal.

Offline information would be stored in the browser through Service Workers and IndexedDB, so users would not need to install separate desktop software.

The software follows the three-layer Major Functional Component Diagram (MFCD) shown in Figure 3.

Figure 3
MFCD



The Presentation Layer contains the Provider Dashboard, Patient Portal, Admin Console, Login/MFA screens, and notifications. These interfaces are built as a React single-page application. The Application Layer sits behind the API gateway and contains the main business logic, including Secure Record Exchange, Identity and Access Management, Patient Record Management, and FHIR/HL7 Integration. The Data Layer contains encrypted medical documents, a PostgreSQL database for users and organizations, a separate FHIR-normalized data store, consent records, and an append-only audit log.

The current PatientPath prototype uses React with Vite and JavaScript for the front end and Node.js with Express for the back end. PostgreSQL 17 serves as the primary relational database, with Prisma 7 used for database access and migrations. Authentication uses Passport.js sessions stored in PostgreSQL, with TOTP-based multi-factor authentication for all users. Role-based access control consists of four roles and 13 permissions enforced through middleware. The prototype currently uses REST APIs, with WebSocket-based real-time updates planned for a later sprint. Docker Compose provides a consistent PostgreSQL development environment for team members, while Jest and scripted RBAC checks support testing. The prototype uses internally FHIR R4-shaped data; full external EHR integration and HL7 v2.x queuing remain outside the current prototype scope.

3 Identification of Case Study

3.1 Target User Group

PatientPath is primarily intended for small-to-mid-sized primary care practices, specialty clinics, and federally qualified health centers (FQHCs) that regularly exchange patient information with outside healthcare organizations. These organizations may have fewer technical resources than large hospital systems while still needing to communicate with hospitals, specialists, and other providers using different electronic health record (EHR) systems. As discussed in Section 1, only 16% of hospitals routinely exchange records with long-term and post-acute care providers outside of their networks, while 71% of office-based physicians work with referral partners that do not have cross-organization electronic exchange capabilities (ASTP/ONC, 2024; ONC, 2022). Rural and lower-resourced healthcare organizations may face additional challenges because of limited IT infrastructure and unreliable internet connectivity.

PatientPath focuses on these organizations because its design allows providers to improve record exchange without replacing their existing EHR systems. The platform acts as a bridge between

systems by converting information into a consistent FHIR R4 format and supporting HL7 v2.x for older systems. Its offline synchronization capabilities are also intended to make important functions available when an organization experiences temporary connectivity problems. These features make rural and lower-resourced healthcare organizations an appropriate group for evaluating whether PatientPath can address the interoperability, referral, and connectivity problems identified in Section 1.

3.2 Case Study Group

The initial PatientPath case study will represent a rural community health network and include participants from the major user roles supported by the platform. The group will include one or two primary care providers responsible for creating referrals, one or two specialists responsible for receiving and reviewing referrals, one administrative staff member responsible for referral coordination and record validation, one system administrator responsible for managing user access and reviewing audit information, and two or three patients who interact with the consent and transfer-status features of the patient portal.

Using these different roles will allow the case study to represent a complete PatientPath referral workflow rather than evaluating only one part of the system. A primary care provider will begin the process by creating a referral and selecting the necessary patient information. The patient will be able to grant permission for the appropriate records to be shared. The receiving specialist will then access the referral and records, while administrative staff can monitor its status and respond to delays. The system administrator will be able to manage access and review activity through the platform's audit information. This allows the case study to follow the referral from its creation through consent, record transmission, specialist receipt, and acknowledgment.

The case study will also allow important PatientPath features to be evaluated under conditions that reflect the problems the product is intended to solve. Referral status tracking can

demonstrate whether users can follow a referral without relying on repeated phone calls or faxes.

Patient consent controls and role-based access can demonstrate how access to records is limited to appropriate users. The prototype's offline capability can also be evaluated by temporarily interrupting connectivity and confirming that supported read-only information remains available through local caching. Full queued-write synchronization and conflict resolution remain part of the real-world product design rather than the current prototype. Together, these activities provide a way to evaluate PatientPath's major features through a realistic healthcare referral scenario.

3.3 Future Expansion

After PatientPath is evaluated with the initial case study group, the platform could be expanded to healthcare organizations that experience similar record-sharing and referral problems.

Potential future users include behavioral health providers, long-term and post-acute care facilities, multi-specialty practices, regional Health Information Exchanges (HIEs), and larger hospital systems that need to exchange information with smaller rural or specialty providers.

These organizations could use the same record-sharing, referral-tracking, consent, and security functions demonstrated by the initial case study.

The platform's EHR-agnostic and plugin-based integration approach is intended to support this expansion without requiring the core PatientPath system to be redesigned for every organization. Additional EHR integrations could be added as the platform grows, while the planned FHIR R4 and HL7 v2.x interoperability layer would provide a consistent method for exchanging information. Beginning with a smaller rural community health network gives the team a focused environment for evaluating PatientPath's main functions before considering deployment across larger and more complex healthcare networks.

4 Glossary

EHR: Electronic Health Record

HIPAA: Health Insurance Portability and Accountability Act

FHIR: Fast Healthcare Interoperability Resources (HL7 standard for health data exchange)

HL7: Health Level Seven (international healthcare data standards organization)

RBAC: Role-Based Access Control

MFA: Multi-Factor Authentication

API: Application Programming Interface

REST: Representational State Transfer (architectural style for APIs)

HIE: Health Information Exchange

PCP: Primary Care Provider

FQHC: Federally Qualified Health Center

CI/CD: Continuous Integration / Continuous Deployment

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